Neurosymbolic AI and Agentic AI

BMED365: Topic List S01–S11

BMED365

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Overview

1. **Fundamental Paradigms**
   - S01: Contrast symbolic and connectionist AI
   - S02: Explain the concept of neurosymbolic integration
2. **Knowledge Graphs and Ontologies**
   - S03: Describe what a knowledge graph is
   - S04: Know about medical ontologies
   - S05: Discuss advantages of neurosymbolic AI in medicine
3. **Case: Neurosymbolic Glioma Diagnosis**
   - S06: Neurosymbolic AI for glioma diagnostics
   - S07: Explain how knowledge graphs can validate predictions
4. **Agentic AI**
   - S08: Define agentic AI and its core characteristics
   - S09: Describe how an AI agent can orchestrate clinical workflows
   - S10–S11: Human-in-the-loop and ethical challenges

S01: Contrast symbolic and connectionist AI

Symbolic AI (GOFAI):

- "Good Old-Fashioned AI"
- Explicit rules and symbols
- Logical reasoning
- Knowledge representation
- Expert systems

Strengths:

- Explainable
- Precise reasoning
- Codifies expert knowledge

Connectionist AI:

- Neural networks, deep learning
- Learns from data
- Subsymbolic representation
- Pattern recognition
- "Black box"

Strengths:

- Learns complex patterns
- Handles noise and uncertainty
- Scales with data

Historical perspective

AI has swung between these paradigms – see AI winters. Now: Can we combine them?

S02: Explain the concept of neurosymbolic integration

Neurosymbolic AI = the best of both worlds

Main idea:

- Combine **neural** (learning from data, pattern recognition)
- with **symbolic** (reasoning, knowledge, explainability)

Integration patterns:

1. **Neural → Symbolic:** Neural network produces symbols for reasoning
2. **Symbolic → Neural:** Symbolic knowledge guides neural network
3. **Hybrid:** Tight integration where both inform each other

Example: Medical diagnosis (brain tumor)

Neural: CNN analyzes MR image, identifies tumor features
Symbolic: Rule-based classification according to WHO criteria
Integrated: Combines image findings with clinical rules for diagnosis

S03: Describe what a knowledge graph is

Knowledge Graph:

- Structured representation of knowledge as a **graph**
- Nodes = **entities** (persons, diseases, drugs)
- Edges = **relations** ("is treated with", "causes")

Triple representation:

(Subject, Predicate, Object) – *(Diabetes, treated_with, Metformin)*

Applications:

- Google Knowledge Graph
- Medical decision support
- Drug interactions

Advantages:

- Machine-readable knowledge
- Reasoning over relations
- Linked Data

Technologies

RDF, OWL, SPARQL, SHACL – standards for knowledge representation, querying and validation

S04: Know about medical ontologies

Ontology = **formal specification of concepts and relations in a domain**

Important medical ontologies/terminologies:

Name	Description	Application
SNOMED CT	Clinical terms, 350,000+ concepts	EHR, classification
ICD-10/11	International disease classification	Reporting, statistics
LOINC	Laboratory investigations	Lab results
RxNorm	Medications	Prescribing
WHO CNS 2021	CNS tumor classification	Glioma diagnosis

WHO 2021 CNS classification

Integrates histological and molecular criteria. Example: IDH-mutant astrocytoma requires both histology **and** IDH mutation status.

S05: Discuss advantages of neurosymbolic AI in medicine

Potential advantages:

1. **Explainability:** Symbolic component provides interpretable reasoning
2. **Data efficiency:** Symbolic knowledge compensates for limited data
3. **Robustness:** Rule-based constraints prevent absurd predictions
4. **Updatability:** New medical knowledge added as rules/facts
5. **Guideline compliance:** Encode WHO criteria, guidelines directly
6. **Validation:** Check if predictions are consistent with known knowledge

Why this is important in medicine

Medicine is a domain where expert knowledge is rich, explainability is critical, and errors can be fatal. Pure deep learning is often insufficient.

S06: Neurosymbolic AI for glioma diagnostics

Case study from Lab 3, Notebook 08:

The problem:

- Glioma classification requires both **image analysis** (MRI) and **molecular markers** (IDH, MGMT, 1p/19q)
- WHO 2021 requires integration of multiple information sources

Neurosymbolic solution:

1. **Neural component:** CNN segments tumor in MRI (BraTS)
2. **Symbolic component:** WHO criteria encoded as knowledge graph
3. **Integration:** Rule-based classification uses CNN output + molecular markers

Result

Diagnosis that is both **image-based** (CNN) and **criteria-compliant** (WHO) – with explainable reasoning.

S07: Explain how knowledge graphs can validate neuro-oncological predictions

Validation of neuro-oncological predictions with symbolic knowledge:

Scenario:

- CNN predicts: "Glioblastoma with IDH mutation"
- Knowledge graph contains: "Glioblastoma is by definition IDH wild-type (WHO 2021)"
- **Conflict detected!**

Validation mechanisms:

1. **Consistency check:** Is the prediction logically consistent?
2. **Rule validation:** Does the prediction satisfy necessary criteria?
3. **Plausibility check:** Is the combination of findings realistic?

Value

Catches potential CNN errors **before** they reach the clinician. Increases trust in the system.

S08: Define agentic AI and its core characteristics

Agentic AI = AI that acts autonomously toward goals

Core characteristics:

1. **Autonomy:** Acts (often iteratively) without continuous human input
2. **Goal-oriented:** Works toward defined goals
3. **Planning:** Breaks down complex goals into subtasks
4. **Tool use:** Can use external tools (APIs, databases)
5. **Reflection:** Evaluates own progress, adjusts strategy
6. **Memory:** Remembers context across multiple interactions and iterations

LLM-based agents:

- LLM as "brain" and "orchestrator" – reasons and plans
- Tool calls (function calling) for actions
- Examples: AutoGPT, LangChain, Claude tool use, OpenAI Operator, CrewAI

S09: Describe how an AI agent can orchestrate clinical workflows

Example: Agentic AI for glioma workup

Scenario: Physician requests "Evaluate new MRI for possible glioma"

The agent can:

1. **Fetch data:** Connect to PACS, download MR images
2. **Analyze:** Run BraTS segmentation, calculate tumor volume
3. **Literature:** Search PubMed for relevant studies
4. **Biobank:** Check if molecular analysis is available
5. **Classify:** Use WHO criteria for tentative diagnosis
6. **Report:** Generate structured report for MDT meeting

Orchestration

The agent **coordinates** multiple AI systems and data sources toward one clinical goal – without the physician having to do each step manually.

S10: Explain the concept of human-in-the-loop in agentic systems

Human-in-the-loop (HITL) in agentic AI:

- Even autonomous agents need **human oversight**
- Especially important for decisions with serious consequences

Implementation patterns:

1. **Approval points:** Agent pauses for approval before critical actions
2. **Confidence threshold:** Human involved when agent is uncertain
3. **Audit trail:** All actions logged for review
4. **Overriding:** Human can interrupt or correct agent

In medical context

Agent can gather data and suggest diagnosis, but the **physician** makes the final decision. The agent is a "super-powered assistant", not an autonomous decision-maker.

S11: Discuss ethical challenges with autonomous AI agents in healthcare

Ethical challenges:

1. **Responsibility:** Who is responsible when an agent makes an error?
   - Developer? Hospital? Physician who "unleashed" the agent?
2. **Autonomy and control:** How much autonomy is too much?
   - Risk of unintended actions
3. **Transparency:** Can we understand why the agent made its decisions?
4. **Security:** Agents with system access = attack surface
5. **Dehumanization:** Risk of reducing human contact in healthcare
6. **Over-reliance:** "Automation bias" – blindly trusting the agent

Principles for responsible agentic AI

Gradual autonomy, robust HITL, clear responsibility allocation, comprehensive testing, continuous monitoring

Summary: Neurosymbolic AI and Agentic AI

Neurosymbolic AI:

- S01–S02: Combines neural (learning) and symbolic (reasoning)
- S03–S04: Knowledge graphs and medical ontologies
- S05–S07: Advantages in medicine, glioma diagnosis case, validation

Agentic AI:

- S08: Autonomous, goal-oriented AI systems with planning
- S09: Orchestration of clinical workflows
- S10–S11: HITL and ethical challenges

Lab 3, Notebook 08

Explore a detailed case study demonstrating neurosymbolic glioma classification and agentic orchestration.